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COURSE SYLLABUS						
Course Code	INTE 401	Course Title	Information Assurance and Security 2		Course Credit	3
Course Description		This course provides an in-depth study of the principles, frameworks, and practices of information assurance and security, focusing on the implementation, management, and auditing of Information Security Management Systems (ISMS) in accordance with ISO/IEC 27001 and related standards such as ISO/IEC 27002, ISO/IEC 27005, NIST SP 800 Series, and COBIT 2019. Students will analyze organizational risks, develop and apply security controls, and create effective policies and incident response strategies. The course emphasizes cryptography, access control mechanisms, digital forensics, vulnerability assessment, and compliance auditing. Emerging topics such as cloud security, IoT vulnerabilities, artificial intelligence (AI) security, and data privacy laws are also discussed to prepare learners for real-world cybersecurity challenges. Through flexible learning activities—including case studies, simulations, hands-on labs, collaborative projects, and reflective learning—students will gain both theoretical understanding and practical competence in protecting and managing information assets within modern digital environments.				
Pre-Requisites		Information Assurance and Security 1	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

As a polytechnic state university, PUP shall develop its students to possess:

1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.
PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO Alignment to ILOs 123456789 Apply knowledge of computing, science, and mathematics appropriate to the discipline. Analyze complex problems, and identify and define the computing requirements appropriate to its solution



	Identify and analyze user needs and take them into account in the selection, creation, evaluation												
	and administration of computer-based systems.												
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.												



	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)		Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Explain and interpret international information security standards (ISO/IEC 27001, NIST, COBIT).								
	Evaluate risks and apply risk management frameworks in information security contexts.								
	Design and implement an Information Security Management System (ISMS) aligned with ISO/IEC 27001.								



	Apply cryptographic principles and access control mechanisms for data protection.								
	Analyze and implement security policies, procedures, and incident response plans.								
	Employ digital forensics and auditing techniques for system assurance and compliance.								
	Utilize emerging security technologies (cloud, IoT, AI security) following best practices.								
	Demonstrate ethical, professional, and legal considerations in cybersecurity management.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	● Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP ● Gain an overview of the course subject and requirements	● Orientation about PUP Vision, Mission, and 10 Pillars ● Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Understand ISO/IEC 27001 structure and concepts	Overview of Information Assurance and Security; ISO/IEC 27001 framework	Lecture, discussion	Reading: ISO 27001 clauses	Quiz 1	Students read ISO/IEC 27001 clauses and				



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						watch short explainer videos.				
3	Explain ISMS concepts and lifecycle	Information Security Management Systems	Case study on ISMS	Watch tutorial videos	Reflection paper	Watch tutorial videos on PDCA cycle; write a				
		(ISMS); PDCA model				short summary on how PDCA ensures continual improvement .				
4	Identify and classify information assets	Asset inventory, classification, and ownership	Workshop: asset mapping.	students identify and categorize assets within simulated organizational contexts.	Activity: Asset register	Read ISO/IEC 27001 Annex A.8 and create an infographic showing asset classification flow.				
5	Develop risk management plan	Risk response strategies, residual risk documentation	Lecture, case analysis	Group meeting	e-Learning module	Virtual collaboration via LMS or Teams to finalize risk register. e-Learning				



						module: Self- paced reading on ISO/IEC 27005 risk process.				
7	Apply cryptographic techniques	Cryptography principles, symmetric/asymmetri c encryption, PKI	Lab demo: encryption tools	Problem- solving session	Watch demo videos	Watch video tutorials on encryption algorithms and digital signatures; complete guided worksheet.				



8-9	Design access control systems	Authentication, authorization, accountability (AAA)	Workshop: access control models	Collaborative session						
9	MIDTERM WRITTEN EXAMINATION									
10	Explain incident response and management	Incident detection, response, and recovery	Incident simulation. Group roleplay							
11	Implement auditing and digital forensics	Audit trails, log management, forensic tools	Demo: forensic imaging	Online Q&A	Reading: Forensics case	Reading of a digital forensics case study; asynchronous Q&A forum discussing ethical evidence handling.				
12	Conduct vulnerability assessment	Penetration testing, vulnerability scanning	Lab exercise	Troubleshooting session	Tool tutorial videos	Watch tool tutorial videos				



						and complete guided worksheet on scanning processes.				
13	Evaluate compliance and governance frameworks	COBIT 2019, ISO 27014, GDPR	Lecture + Group activity	Discussion	Reading module	Reading module on COBIT and GDPR; asynchronous forum discussion on data governance.				



14	Discuss security in cloud and IoT environments	Cloud security architecture, IoT risk mitigation	Seminar	Expert talk	Watch AWS/Azure security videos	Watch AWS and Azure security videos; reflection post on IoT risk management.				
15	Examine emerging threats and AI security	AI-based threats, ML security, zero-trust frameworks	Case study	Open forum	Read IEEE whitepaper	Read IEEE whitepaper on AI security; asynchronous discussion board participation.				
16	Apply ethical and legal principles	Cyber laws, data privacy, professional ethics	Debate session	Consultation	Reading on privacy acts	Reading on Data Privacy				
						Acts and related laws; reflection				



						n journal submiss io n.				
17	Integrate ISMS in organizational context Present and evaluate ISMS design project	ISMS documentation, audit preparation ISMS design and implementation presentation	Capstone workshop Final presentation Feedback session Peer review	Independe nt project work Submission of final report	Project draft Final Project (CLO3, CLO6)	Indepen d ent project work and peer review using online collabor ati on platform s.				
18	FINAL WRITTEN EXAMINATION									



1 8	FINAL EXAMINATION									

LIST OF REFERENCES

BOOKS:

WEBSITE:

Kennesaw State University. (n.d.). *Model curriculum for information security/cybersecurity*. Kennesaw State University. <https://www.kennesaw.edu/coles/centers/cyber-center/resources/docs/model-curriculum-2021.pdf>

University of Hawaii. (2019). *ISA 400: Management of information security [Course syllabus]*. University of Hawaii. https://www2.hawaii.edu/~probbins/UHWO.ISA400.SP19/0%20-%20ISA400%20SYLLABUS%20_Robbins%20Spring%202019_.pdf

University of British Columbia, School of Information. (2023). *IT security, information assurance, and risk management [Course syllabus]*. University of British Columbia. <https://ischool.ubc.ca/wp-content/uploads/sites/46/2023/03/arst-575j-libr-514k-syllabus.pdf>

Certible. (2024). *ISO/IEC 27001 lead auditor syllabus (CCSI009LA Syllabus v2024.14)*. Certible. https://www.certible.com/assets/static/CCSI009LA_Syllabus_v2024.14_en.pdf

IT Governance. (n.d.). *Information security & ISO 27001 [Course outline]*. IT Governance Ltd. https://www.itgovernance.co.uk/files/infosec_101v1.1.pdf

IT Governance USA. (n.d.). *Certified ISO 27001:2022 ISMS foundation training course*. IT Governance USA. <https://www.itgovernanceusa.com/shop/product/certified-iso-270012022-isms-foundation-training-course>



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2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION



- | | | |
|------------------------|------------------------|--|
| ★★★★★ | ★★★★★ | ★★★★★ |
| • Employability | • Teaching | • Social Responsibility |
| • Online Learning | • Academic Development | • Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND) |
| ★ | • Inclusiveness | |
| • Internationalization | | |





Advisera. (n.d.). *Free ISO 27001 training courses & certification*. Advisera. <https://advisera.com/training/iso-27001-courses/>

Intertek Academy North America. (n.d.). *Implementing an ISMS (ISO/IEC 27001:2022)*. Intertek Academy. <https://learning.intertek.com/course/information-security/implementing-an-isms-iso-iec-270012022/>

PECB. (n.d.). *ISO/IEC 27001 lead implementer / lead auditor training*. PECB. <https://pecb.com/en/education-and-certification-for-individuals/iso-iec-27001>

CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to- face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is “INC” grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don’t tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student’s own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be

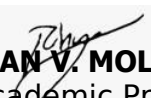
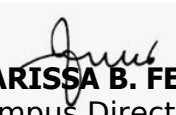
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	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.			
COURSE REQUIREMENT/S	GRADING SYSTEM			
Students are required to develop an Information Security Management System (ISMS) implementation plan, perform risk assessments, and create security policies aligned with ISO/IEC 27001 standards. They must complete hands-on cybersecurity laboratory exercises, participate in class discussions and presentations, and submit a final ISMS project with documentation and defense demonstrating the integration of learned concepts and practical applications.	Mid Term		Final Term	
	Class Standing	70%	Class Standing	70%
	Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper		Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper	
	Midterm Examinations	30%	Final Examination ()	30%
		100%		100%
	Final Grade = (Midterm Grade + Final Term Grade) / 2			
CLASS INFORMATION		FACULTY INFORMATION		
Course, Year and Section: BSIT-TG- 4-1		Name of Faculty: MODESTO, LADY MELINDA MINETTE LOPEZ		
Day and Time: F 05:00PM-08:00PM		Consultation Time:		
Room: TGA403		Office Tel. No./ Mobile Phone No.		



School Year:	Semester: First Semester	E-Mail Address:
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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Faculty Member	 DR. RHYAN V. MOLINAR Head of Academic Programs	 DR. MARISSA B. FERRER Campus Director	 PROF. PASCUALITO B. GATAN Vice President for Campuses